**13.Perform Affine Transformation on the given image using python and Open CV**

Aim

To write and execute a Python program to perform affine transformation on an image using OpenCV.

Procedure

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Get the height and width of the image.
4. Define three points in the original image.
5. Define three corresponding points in the transformed image.
6. Calculate the affine transformation matrix using cv2.getAffineTransform().
7. Apply the transformation using cv2.warpAffine().
8. Display the original and transformed images.

Code

```
import cv2
import numpy as np

img = cv2.imread("sample.jpg")

if img is None:
    print("Image not found")
else:
    rows, cols = img.shape[:2]

    pts1 = np.float32([
        [50, 50],
        [200, 50],
        [50, 200]
    ])

    pts2 = np.float32([
```

```
    [10, 100],
    [200, 50],
    [100, 250]
])

matrix = cv2.getAffineTransform(pts1, pts2)

result = cv2.warpAffine(img, matrix, (cols, rows))

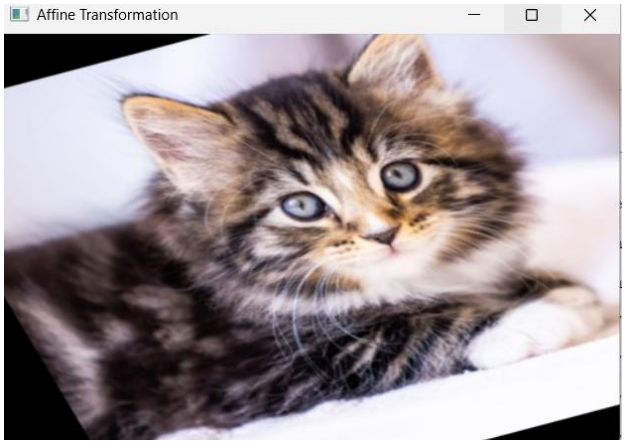
cv2.imshow("Original Image", img)
cv2.imshow("Affine Transformation", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

SAMPLE INPUT:




SAMPLE OUTPUT:




**Result**

**Thus, affine transformation was successfully performed on the given image using Python and OpenCV.**